FORAM FANASIA

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SUMMARY

Data Scientist with experience in quantitative modeling, empirical research, and large-scale data analysis. Skilled in Python and SQL for developing data pipelines, testing hypotheses, and extracting signals from complex datasets, with a strong focus on systematic modeling and decision making.

WORK EXPERIENCE

University of Massachusetts Dartmouth — Dartmouth, Massachusetts
Research Associate — *July 2024 - present*

- Built analytical workflows using Python and Excel to process **10,000+ data points across 15+ test conditions**, improving the accuracy of data analysis by ~30% and enabling the structured evaluation of system performance across multiple experimental scenarios
- Designed structured datasets across **5+ configurations and 20+ experimental runs**, reducing preprocessing time by ~40% and enabling consistent comparison of results across varying conditions and analytical requirements
- Applied statistical analysis and regression across datasets spanning **Re 3,000–30,000**, improving accuracy of trend identification by ~15% and enabling a deeper understanding of key performance drivers in complex systems
- Automated data validation and reporting workflows using Python, reducing manual effort by ~35% and improving consistency and reliability of analytical outputs across recurring evaluations and research reporting
- Collaborated with **3+ research teams and contributed to 2 publications**, improving workflow coordination by ~20% and ensuring timely delivery of structured analytical insights across multiple projects

University of Massachusetts Dartmouth — New Bedford, Massachusetts
Data Scientist — *Sep 2023 - June 2024*

- Developed simulation models using Python and Julia, generating **GB-scale datasets across 50+ runs**, enabling structured data analysis and improving the efficiency of model evaluation processes by ~25% across projects
- Engineered data pipelines using Python and SQL, reducing data processing time by ~35% and enabling repeatable workflows for large dataset analysis across multiple simulation scenarios and parameter variations. Applied statistical and machine learning techniques on multi-year datasets, improving predictive insights by ~12% and enabling identification of key trends across complex data environments
- Built **20+ visualization tools and reports**, improving clarity of analytical findings and reducing interpretation time for stakeholders by ~30% across technical and research teams. Managed 3–5 concurrent analytical projects, improving delivery timelines by ~25% while maintaining high-quality outputs across modeling, analysis, and reporting workflows

Freie University — Berlin, Germany
Data Analyst — *May 2022 - July 2023*

- Analyzed **ERA5 and CMIP6 datasets (>100GB, 30+ years)**, improving efficiency of large-scale data analysis by ~30% and enabling structured evaluation of environmental and system-level trends
- Built data pipelines processing **10,000+ grid points across regions**, improving workflow efficiency by ~35% and enabling scalable analysis across spatial and temporal datasets
- Applied time-series and statistical techniques to multi-year datasets, improving trend detection accuracy by ~15% and enabling stronger insights into system behavior
- Developed **25+ visualizations and reports**, reducing reporting complexity by ~20% and improving communication of analytical results across interdisciplinary teams
- Collaborated with **2+ international teams**, improving coordination efficiency by ~20% and ensuring consistent delivery of high-quality analytical outputs across large-scale projects

PROJECT

Multi-Asset Dual-Momentum Index: Backtesting, Risk Simulation & Cost Analysis

- Engineered a dual-momentum multi-asset backtesting system in Python (SPY, TLT, GLD, QQQ, EEM) with a **SQLite persistence layer across 6 tables** (prices, weights, trades, performance, liquidity, Monte Carlo results), covering **10 years and 100+ monthly rebalance periods**
- Designed a signal framework combining a **blended 3/6/12-month absolute-momentum gate** with 12-month relative-momentum ranking and inverse-volatility weighting, plus a scored defensive-basket fallback (bonds, gold, cash) to reduce reliance on any single "safe" asset during regime shifts
- Quantified real-world frictions by modeling **transaction costs and liquidity constraints** (participation rate vs. 20-day average daily volume) on a $10M notional book, isolating a ~2.8% cost drag on total return over the backtest period
- Built a vectorized, batched **block-bootstrap Monte Carlo engine scaled to 1,000,000 simulated paths** for VaR/CVaR and drawdown-distribution estimation, cutting runtime from an hours-long per-path loop to under a minute
- Ran a parameter sensitivity sweep against real 2015–2024 price history to isolate the highest-impact lever, then validated the resulting fix, **cutting maximum drawdown to -22% vs. -34% for SPY** while holding a comparable Sharpe ratio (~0.65 vs. 0.67), and published the system to a version-controlled GitHub repository

EDUCATION

University of Massachusetts Dartmouth — Dartmouth, Massachusetts
Master of Mechanical Engineering — *June 2026*

National Institute of Technology — Surat, India
Integrated Master's of Physics — *June 2023*

SKILLS

Programming Languages: Python, SQL, MATLAB, Julia, R

Technologies & Tools: Quantitative Research, Statistical Analysis, Machine Learning, Data Pipelines, Large Dataset Processing, Empirical Research, Hypothesis Testing, Optimization, Data Visualization, Git

Libraries and Frameworks: Pandas, NumPy, SciPy, scikit-learn, Statsmodels, Matplotlib, Seaborn